

Visual Cognition reconstruction using a single-channel EEG

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Abstract

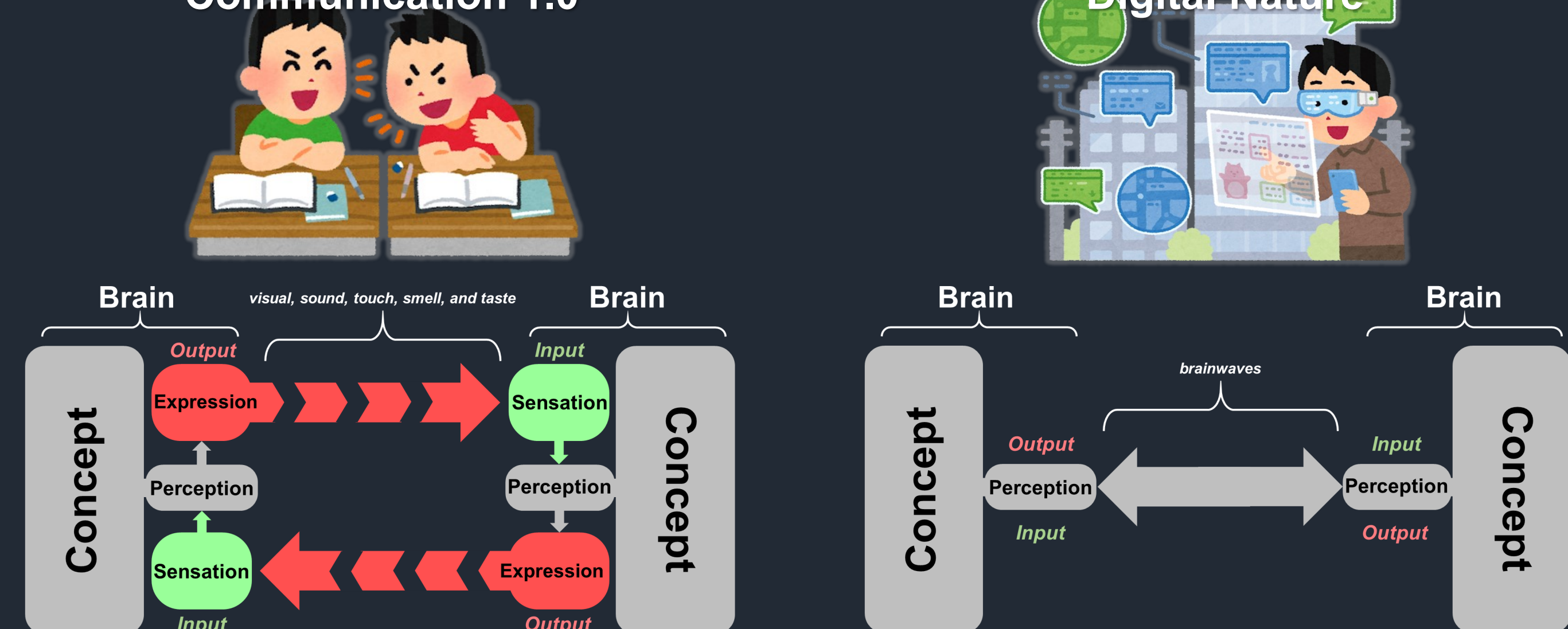
Real-time decoding of brain activity by brain-computer interface has not been fully realized despite many efforts in the field. Kamitani and others have succeeded in visualizing fMRI signal as an intermediate layer of a convolutional neural network to decode what is being seen. Although the spatial resolution of fMRI is of the highest caliber, it is limited by high cost and operational skill requirement. Another method that is widely used to measure and record brain activity is via electroencephalogram (EEG). It can detect different frequencies of waves from brain signals. Compared to fMRI, EEG may be inferior in terms of spatial resolution, but it does have advantages over fMRI. This includes better temporal resolution, low-cost, easier to use, and some are even portable. These features allow data collection from large number of individuals possible.

To achieve higher spatial resolution in EEG, more channels are opted as a dense network of electrodes creates a spatial map to a degree. However, this results to higher cost, decreased ease-of-use and comfort. As firm believers that we primarily need to address the lack of data in brain imaging, we used single-channel EEG headsets which are powerful yet underexploited tools in neuroscience. This follows the ideology of bigger quantity over quality. In this study, we attempted to reconstruct images based on the detected brain signals from 15 healthy subjects. This was done by creating a library of the images and the associated brain waves that were detected upon exposure to the image and during image recalling phase.

Vision

Revolutionize the way humans **shape** and **interact** with their environment.

Communication 1.0



Re-invent communication, not limited by speech, language difference, or disabilities; to have a world where we can interact and express ourselves (to people and devices) using the signals generated by our brain.

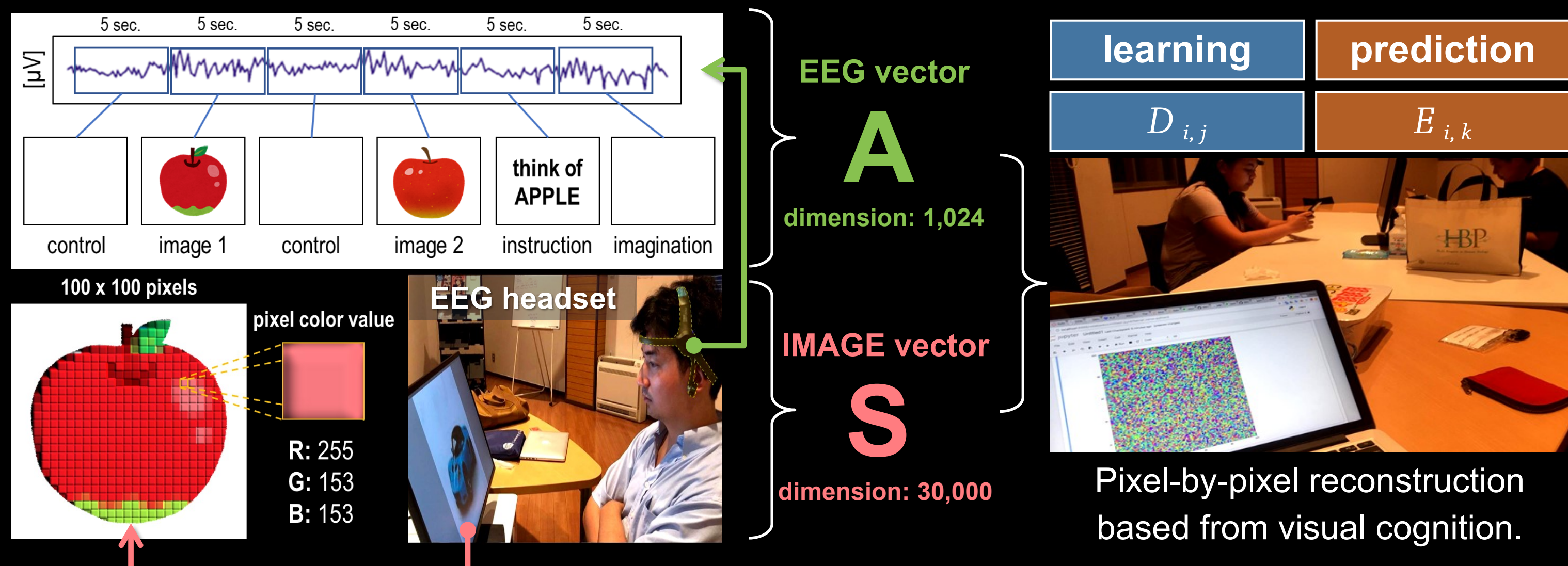
Experimental Design (EEG + AI)

The recent advances in brain activity monitoring and machine learning have allowed us to understand the mechanism of visual cognition and its application to brain-based communication.



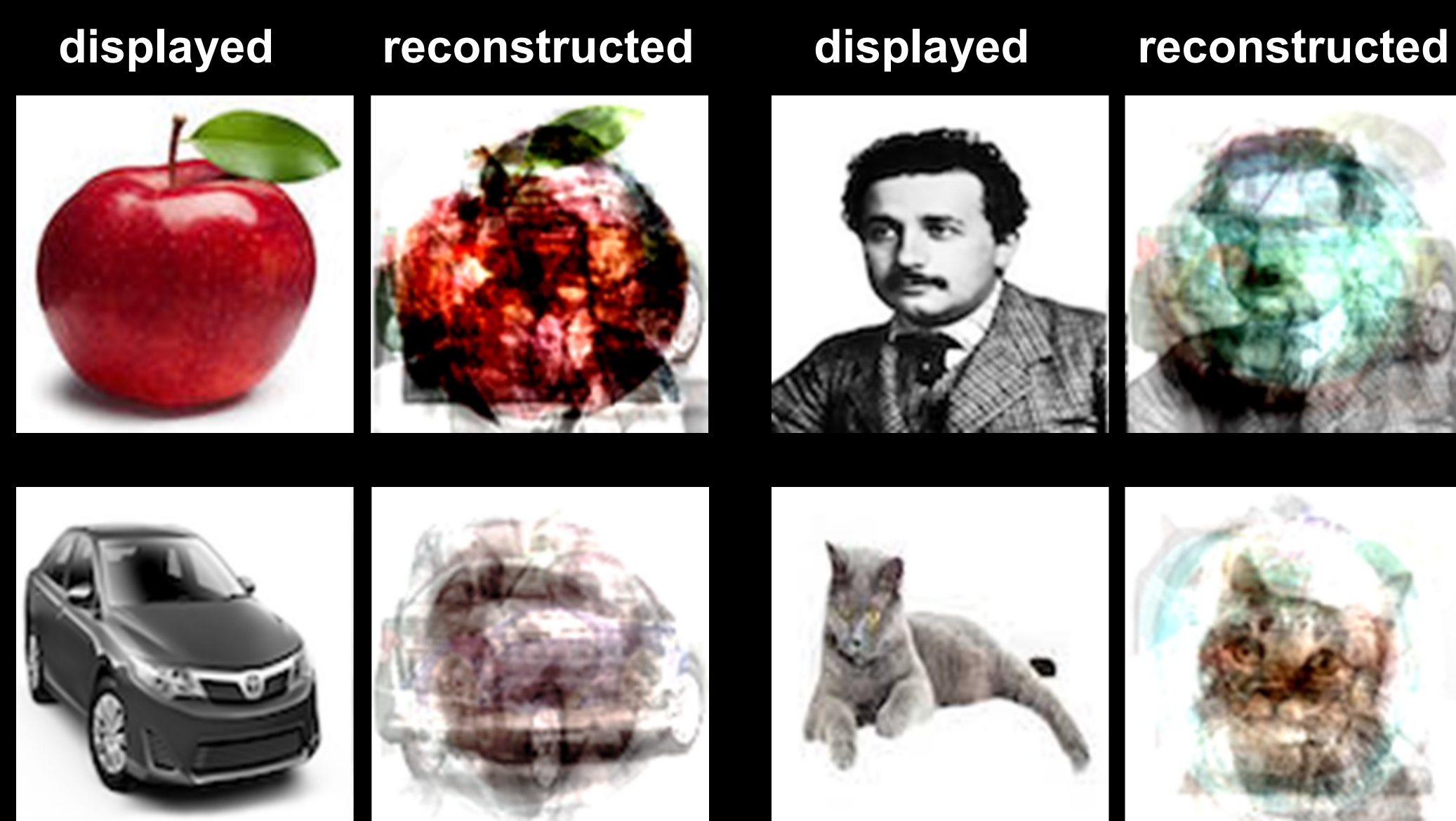
Issue: **Expensive** devices and **limited** learning systems

Four (4) object categories having 10 images each was used in learning the relationship between the **EEG** and **IMAGE** vectors. The prediction algorithm was generated from the data of 15 volunteers.



Results and Perspectives

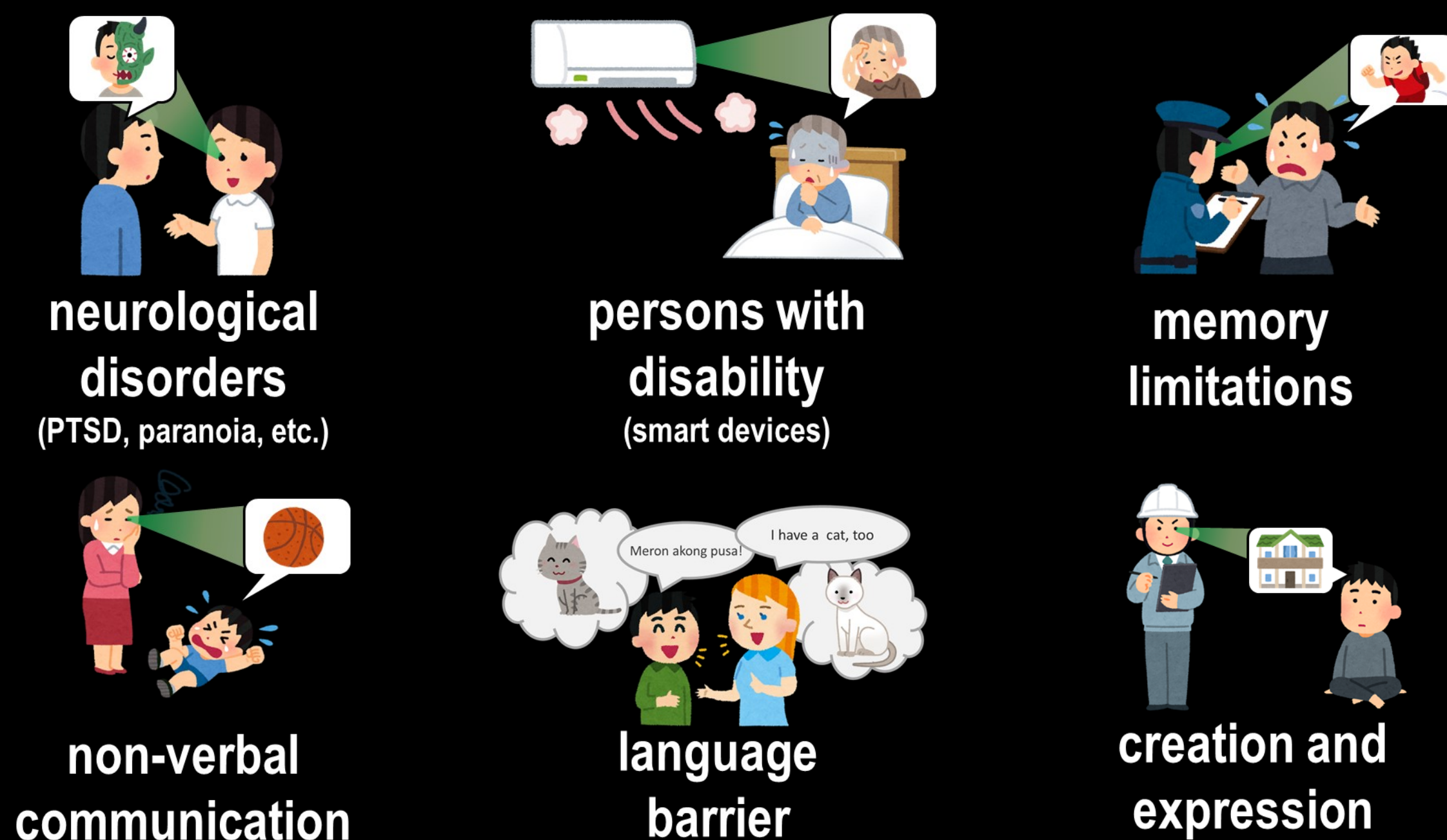
We have previously used single-channel EEG signal to direct user's attention and allow them to choose from a set of predetermined commands. This system was designed as a communication device for patients suffering from disability. On the other hand, using single-channel EEG for a complex task of visual cognition reconstruction is **first of its kind**.



Despite its limited resolution, its **simple setup** allowed us to conduct several trials that could be used for constructing the library. Interestingly, the system can easily distinguish the displayed image among different categories. Reconstruction obtained higher accuracy in simple shapes and color (apples) and images that have a face which are both consistent with previous reports. This study shows the capability of low-resolution wearable EEG device for brain-based communication.

Expanding Human Capabilities

Visual cognition comprises 80% of the information we perceive. Imagine all the possibilities we can do if we can reconstruct these thoughts in to digital form. This will change the way we look on:



We see with our brains, not with our eyes.
我々は目ではなく脳で見ている。

- Paul Bach-y-Rita
Neuroscientist
(1934-2006)

